UNIT IV

Cloud computing challenges: Economics of the cloud, cloud interoperability and standards, scalability and fault tolerance, energy efficiency in clouds, federated clouds, cloud computing security, fundamentals of computer security, cloud security architecture, cloud shared responsibility model, security in cloud deployment models.

❖ Economics of Cloud

Economics of Cloud Computing is based on the PAY AS YOU GO method. Users/Customers must have to pay only for their way of the usage of the cloud services. It is definitely beneficial for the users. So, the Cloud is economically very convenient for all. Another side is to eliminate some indirect costs which is generated by assets such as license of the software and their support. In the cloud, users can use software applications on a subscription basis without any cost because the property of the software providing service remains to the cloud provider.

Economical background of the cloud is more useful for developers in the following ways:

- Pay as you go model offered by cloud providers.
- Scalable and Simple.

In particular, cloud computing allows:

1. Reducing the capital costs associated to the IT infrastructure
2. Eliminating the depreciation or lifetime costs associated with IT capital assets
3. Replacing software licensing with subscriptions
4. Cutting the maintenance and administrative costs of IT resources

Pricing Strategies:

There are three different Pricing Strategies that are introduced by Cloud Computing: Tiered Pricing, Per-unit Pricing, and Subscription-based Pricing. These are explained as following below.

***a) Tiered pricing**:* In this model, cloud services are offered in several tiers, each of which offers a fixed computing specification and SLA(service level agreement) at a specific price per unit of time.

This model is used by Amazon for pricing the EC2 service, which makes available different server configurations in terms of computing capacity (CPU type and speed, memory) that have different costs per hour.

b) Per-unit pricing: This model is more suitable to cases where the principal

source of revenue for the cloud provider is determined in terms of units of specific services, such as data transfer and memory allocation. In this scenario customers can configure their systems more efficiently according to the application needs. This model is used, for example, by GoGrid, which makes customers pay according to RAM/hour units for the servers deployed in the cloud.

c)Subscription-based pricing: In this model, users are paying periodic subscription fees for the usage of the software.

❖ Cloud Interoperability and Standards

1. Introduction

Cloud computing provides on-demand services such as storage, processing, and networking over the internet. As organizations increasingly adopt multi-cloud and hybrid cloud environments, it becomes essential that different cloud systems can work together efficiently.

This requirement is fulfilled by:

- Cloud Interoperability → Ensures systems can interact
- Cloud Standards → Provide rules for compatibility

2. Cloud Interoperability

Definition

Cloud interoperability is the ability of different cloud platforms, applications, and services to communicate, exchange data, and function together seamlessly without manual intervention or conversion. It ensures that systems work as an integrated whole despite being built on different technologies or managed by different providers.

Key Characteristics

- Enables integration across multiple cloud providers
- Supports data sharing and migration
- Promotes flexibility and scalability
- Reduces vendor lock-in

Types of Cloud Interoperability

2.1 Application Interoperability: Application Interoperability refers to the ability of applications running on different cloud platforms to communicate, exchange information, and interact with each other seamlessly.

How It Works

Applications connect through well-defined communication protocols and interfaces

Uses standardized web services to exchange data

Enables hybrid and multi-cloud application architectures

Real-World Example

An e-commerce application hosted on AWS can seamlessly interact with a payment processing service on Azure and a database on Google Cloud. They communicate through REST APIs without knowing details of each other's infrastructure.

2.2 Data Interoperability: Data interoperability ensures that data stored in different cloud platforms can be easily shared, accessed, and understood across systems without loss of meaning or format compatibility issues.

Why Data Interoperability Matters

- Enables data migration between cloud providers without restructuring
- Prevents data loss, corruption, or incompatibility during transfers.
- Allows organizations to access data regardless of where it's stored
- Supports business continuity and disaster recovery strategies

Example

A company stores customer data in AWS in JSON format. They can easily migrate this data to Azure or Google Cloud because JSON is a universal standard. Without standardized formats, data would be unreadable in the new system.

2.3 Platform Interoperability:

Platform interoperability allows applications and workloads to run across different cloud platforms and infrastructure providers without modification or recompilation.

How It Works

Applications are abstracted from underlying infrastructure using virtualization and containerization

The same application package can be deployed on different cloud environments

Real-World Example

A web application containerizedth Docker can be deployed on AWS ECS, Azure Kubernetes Service, or Google Cloud Kuberrietes Engine without code changes. The container ensures consistent behaviour across all platforms.

Advantages of Interoperability

- Avoids vendor lock-in
- Enables multi-cloud strategies
- Improves resource utilization
- Enhances business continuity
- Supports innovation and flexibility

Challenges of Interoperability

- Proprietary APIs by cloud providers
- Lack of universal standards
- Security and privacy concerns
- Data format inconsistencies
- Complex integration processes

3. Cloud Standards

Definition

Cloud standards are a set of rules, protocols, and guidelines that ensure interoperability, portability, and compatibility among cloud systems. They define common specifications for cloud services, APIs, data formats, security, and management. Standards act as the foundation enabling different systems to communicate and work together.

Types of Cloud Standards

3.1 *Open Standards*

Characteristics

Publicly available and freely accessible to everyone

Developed by industry consortiums or standards bodies

Anyone can implement and use these standards

Promote interoperability and prevent vendor lock-in

Examples of Open Standards

HTTP/HTTPS (HyperText Transfer Protocol):

Foundation of web communication. Enables data transfer between servers and clients.

REST APIs (Representational State Transfer):

Architectural style for web services. Uses HTTP methods (GET, POST, PUT, DELETE) for operations.

JSON and XML:

Universal data formats for information exchange.

3.2 Proprietary Standards

Characteristics

Developed by individual vendors for their products
Not publicly standardized or universally available
May restrict interoperability and increase complexity
Can lead to vendor lock-in

Examples

AWS-specific APIs that don't work on Azure or Google Cloud
Microsoft Azure proprietary authentication mechanisms
Vendor-specific database query languages

Open vs Proprietary: Comparison

Aspect	Open Standards	Proprietary Standards
Availability	Publicly available	Restricted access
Interoperability	High	Low
Vendor Lock-in	Prevents lock-in	Causes lock-in
Development	Community-driven	Vendor-driven

Standard Organizations

NIST (National Institute of Standards and Technology)

U.S. government agency responsible for developing cloud computing standards

Published NIST SP 800-145 defining cloud computing characteristics

Focuses on security, interoperability, and cloud architecture

ISO/IEC (International Organization for Standardization)

International body developing worldwide standards

ISO/IEC 27001: Information security management

ISO/IEC 22623: Cloud computing security requirements

Open Grid Forum (OGF)

Industry organization developing grid and cloud computing standards

Focuses on interoperability and portability standards

<u>Important Cloud Standards</u>

1. OVF (Open Virtualization Format)

OVF is an open standard for packaging, distributing, and deploying virtual machines and applications across different cloud platforms and hypervisors.

- Ensures VMs created on one platform can run on another without conversion
- Includes VM.configuration, hardware requirements, and metadata
- Supports multi-cloud portability

Example

A company creates a virtual machine in OVF format. This single file can be deployed on VMware vSphere, Microsoft Hyper-V, or KVM without modification

2. CDMI (Cloud Data Management interface)

CDMI is a standard interface for managing data stored in cloud storage systems. It defines how to access, manipulate, and manage cloud data consistently

- Enables uniform data management across different cloud storage providers
- Supports data querying, retrieval, and metadata management
- Promotes data interoperability

Example

Using CDMI, an organization can store data on AWS S3 and manage it using the same commands and procedures used for Azure Blob Storage, ensuring consistency.

3. OAuth 2.0 (Authorization Framework)

OAuth 2.0 is an open standard authorization protocol that allows users to grant applications access to their resources without sharing passwords.

- Enables secure authentication across cloud platforms
- Allows users to use single credentials across multiple services
- Improves secunty by eliminating password sharing

Example

"Login with Google" buttons on applications use OAuth 2.0. Users authenticate with Google once and grant apps permission to access their data without sharing their Googte password.

4. SLA (Service Level Agreement)

SLA is a formal agreement between cloud providers and customers defining service guarantees, performance metrics, and remedies for failures.

- Defines uptime guarantees (e.g., 99.99% availability)
- Specifies performance metrics (response time, throughput)
- Provides compensation for service failures
- Ensures accountability and reliability

Example

AWS guarantees 99.99% uptime for EC2 services. If actual uptime falls below this, AWS provides service credits as compensation.

5. OpenStack APIs

OpenStack is an open-source cloud computing platform that provides standardized APIs for cloud infrastructure. It serves as both a reference implementation and a set of open standards

- Offers alternative to proprietary cloud platforms
- Provides open standards for compute, storage, and networking
- Enables private cloud deployments
- APIs can be adopted by other cloud platforms

Example

Organizations can build private clouds using OpenStack, with APIs compatible with major pub clouds, reducing vendor lock-in risks.

❖ Scalability and Fault Tolerance

Scalability:

Scalability is the capability of a cloud system to handle increasing or decreasing workload by adding or removing resources dynamically.

Types of Scalability

1. Vertical Scalability (Scale Up/Down)

- Enhancing the capacity of a single server
- Involves adding CPU, RAM, storage
- Simple but has hardware limits

2. Horizontal Scalability (Scale Out/In)

- Adding multiple servers to distribute workload
- Highly flexible and widely used in cloud
- Provides better fault handling

Characteristics of Scalability

- **Elasticity** – automatic adjustment of resources
- **Responsiveness** – quick reaction to demand changes
- **Resource pooling** – shared resources among users
- **Measured service** – usage is monitored and billed

Mechanisms Used

- **Auto Scaling**: Automatically adjusts computing resources by adding or removing servers based on demand. It ensures optimal performance and avoids resource wastage during low or high traffic.
- **Load Balancing** – Distributes incoming network traffic evenly across multiple servers. This prevents overload on a single server and improves system performance and reliability.
- **Distributed Systems** – A distributed system consists of multiple interconnected computers working together as one system. It improves scalability, fault tolerance, and resource sharing across different locations.

Advantages

- Efficient handling of peak loads
- Improved application performance
- Cost optimization (no resource wastage)

- Supports business growth

Limitations

- Complex system design
- Requires proper monitoring
- Vertical scaling has physical limits

Real-Time Example

- Streaming platforms increase server capacity during live events to handle millions of users.

Fault Tolerance:

Fault tolerance is the ability of a system to continue its operation without interruption even when one or more components fail.

Key Concepts

- **Failure Detection** – identifying failures quickly
- **Recovery Mechanism** – restoring system functionality
- **Redundancy** – having extra components as backup

Techniques

1. Redundancy

- Duplicate hardware or software components are maintained as backups.
- Ensures system continues working even if one component fails.

2. Data Replication

- Data is stored in multiple geographic locations or servers.
- Prevents data loss and ensures availability during failures.

3. Failover Systems

- Automatically switches to a backup system when the main system fails.
- Ensures continuous service with minimal or no downtime.

4. Load Balancing

- Distributes workload evenly across multiple servers.
- Prevents overloading and improves performance and reliability.

5. Checkpointing

- Saves the current state of a system at regular intervals.

- Helps in quick recovery by restarting from the last saved state.

Characteristics

- High availability (24/7 service)
- Reliability
- Disaster recovery capability
- Minimal downtime

Advantages

- Ensures continuous service
- Prevents data loss
- Enhances user trust
- Supports critical applications (banking, healthcare)

Limitations

- Increased cost due to redundancy
- Complex implementation
- Requires proper maintenance

Real-Time Example

- Banking systems use multiple servers so that even if one fails, transactions continue without interruption.

❖ Energy Efficiency in Clouds

Energy efficiency in cloud computing refers to the **efficient use of electrical power in data centers, servers, storage, and network infrastructure** to perform computing tasks while minimizing energy consumption. It is an important aspect of **green computing** and sustainable IT operations.

Why Energy Efficiency is important?

Cloud data centers consume **massive amounts of electricity** due to:

- Continuous server operation
- Cooling systems
- Storage and networking devices

Improving energy efficiency helps:

- Reduce operational cost
- Lower carbon emissions
- Improve system performance
- Protect the environment

Techniques to Improve Energy Efficiency

1. Virtualization

Virtualization allows multiple virtual machines (VMs) to run on a single physical server, increasing resource utilization and reducing the number of active servers.

Benefit: Fewer servers → Less power consumption

2. Server Consolidation

Workloads from under-utilized servers are combined into fewer machines so that idle servers can be turned off or put into sleep mode.

3. Dynamic Resource Allocation

Cloud systems allocate CPU, memory, and storage resources only when required and release them when demand decreases.

4. Auto Scaling

Automatically increases or decreases the number of servers based on user demand, preventing unnecessary energy usage during low traffic.

5. Energy-Efficient Hardware

Using modern processors, SSDs, and low-power memory reduces electricity consumption while maintaining performance.

6. Efficient Cooling Techniques

Cooling systems consume nearly **40–50%** of data center energy. Techniques include:

- Liquid cooling
- Free air cooling

- Hot and cold aisle containment

7. Power-Aware Scheduling

Tasks are scheduled to run on fewer servers during low load so that other servers can be switched off to save energy.

8. Use of Renewable Energy

Cloud providers use solar, wind, and hydroelectric power to operate data centers, reducing environmental impact.

Advantages of Energy-Efficient Cloud Computing

- Reduced electricity consumption
- Lower operational and maintenance cost
- Reduced greenhouse gas emissions
- Improved sustainability and green IT
- Better resource utilization

Challenges

- High initial cost of energy-efficient infrastructure
- Complex management of virtual machines and workloads
- Maintaining performance while reducing power usage

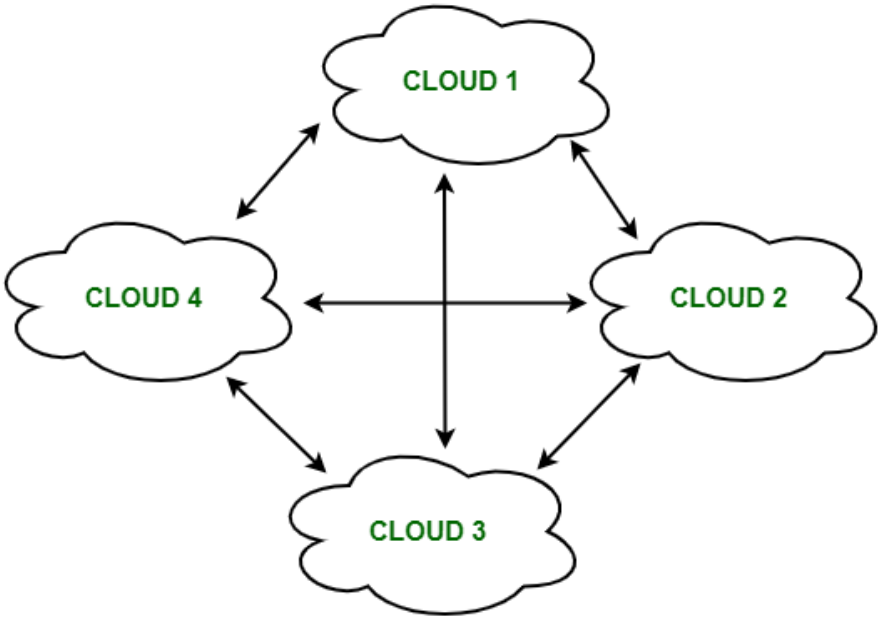
Real-World Example

Large cloud providers like AWS, Google Cloud, and Microsoft Azure build energy-efficient data centers that use renewable energy and advanced cooling to reduce power consumption.

❖ Federated Clouds

Federated clouds refer to a cloud computing model in which multiple independent cloud service providers or data centers are interconnected to share resources and services. These clouds collaborate with each other while maintaining their individual autonomy, forming a unified and scalable computing environment. The main goal of federated clouds is to enable flexibility, improve resource utilization, and avoid dependency on a single cloud provider.

In a federated cloud environment, different clouds communicate through standardized interfaces and protocols. This interoperability allows users to access computing resources such as storage, processing power, and applications from multiple cloud platforms seamlessly. For example, an organization may use a private cloud for sensitive data and integrate it with public clouds like AWS or Microsoft Azure for additional computing power during peak demand.



<u>The architecture of Federated Cloud:</u>

The architecture of Federated Cloud consists of three basic components:

1. Cloud Exchange

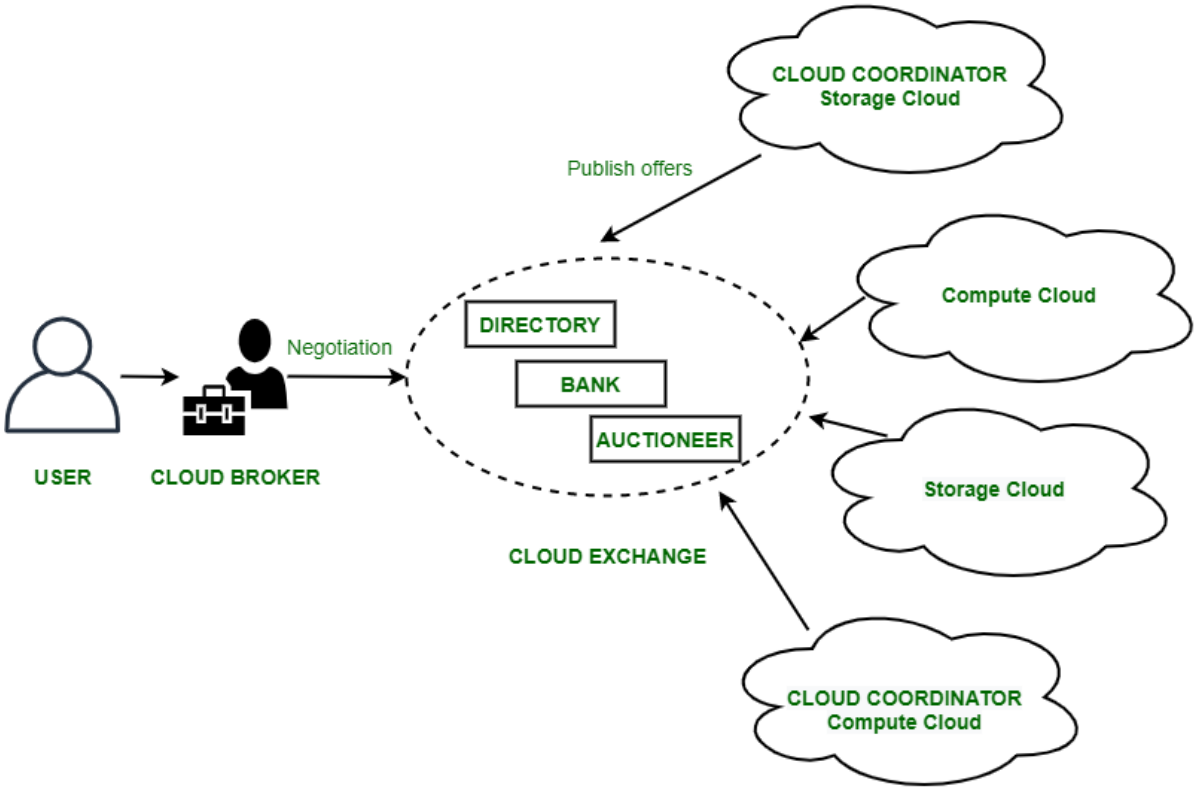
The Cloud Exchange acts as a mediator between cloud coordinator and cloud broker. The demands of the cloud broker are mapped by the cloud exchange to the available services provided by the cloud coordinator. The cloud exchange has a track record of what is the present cost, demand patterns, and available cloud providers, and this information is periodically reformed by the cloud coordinator.

2. Cloud Coordinator

The cloud coordinator assigns the resources of the cloud to the remote users based on the quality of service they demand and the credits they have in the cloud bank. The cloud enterprises and their membership are managed by the cloud controller.

3. Cloud Broker

The cloud broker interacts with the cloud coordinator, analyzes the Service-level agreement and the resources offered by several cloud providers in cloud exchange. Cloud broker finalizes the most suitable deal for their client.



Properties of Federated Cloud

1. Interaction Types
2. Visibility
3. Monitoring
4. Service Publishing
5. Resource Sharing

Benefits of Federated Cloud

- Reduces **energy consumption**
- Increases **reliability**
- Reduces **cost and time** using dynamic scaling
- Enables **easy scalability**

Challenges of Federated Cloud

- Load distribution problem
- Interoperability issues
- Provider selection difficulty
- Complex environment

Technologies in Federated Cloud

1. OpenNebula

- Manages distributed data centers
- Supports different systems (heterogeneous environments)

2. Aneka Coordinator

- Helps different clouds communicate
- Improves performance and connectivity

3. Eucalyptus

- Open-source cloud platform
- Manages **compute, storage, and network resources**
- Supports scaling up/down based on demand

❖ Cloud Computing Security

Cloud computing security refers to the set of policies, technologies, and controls used to protect data, applications, and services stored in cloud environments. As organizations increasingly rely on cloud computing for storing sensitive information and running critical applications, ensuring security has become a major concern.

One of the primary aspects of cloud security is data protection. Data stored in the cloud must be safeguarded from unauthorized access, breaches, and loss. This is achieved through techniques such as encryption, where data is converted into a secure format that can only be accessed with proper authorization. Both data at rest (stored data) and data in transit (data being transferred) need to be protected.

Another important aspect is identity and access management (IAM). It ensures that only authorized users can access specific resources in the cloud. Strong authentication methods, such as multi-factor authentication, are used to verify user identities. Proper access control mechanisms help prevent misuse of resources and protect sensitive information.

- **Cloud Deployment Models**

Cloud deployment models define how cloud infrastructure is owned, managed, and made available to users. There are 4 cloud deployment models:

- **Public Cloud:** Shared infrastructure available to the public.
- **Private Cloud:** Dedicated cloud for a single organization.
- **Hybrid Cloud:** Combination of public and private clouds.
- **Community Cloud:** Shared by organizations with similar requirements.

- **Security in Cloud Computing**

Cloud security involves protecting cloud environments, data, and applications from unauthorized access, malware, DDoS attacks, and data breaches.

- Ensures both internal and external security.
- Protects sensitive organizational data.

- **Planning of Security in Cloud Computing**
  - Identify which resources can be moved to the cloud.
  - Assess data sensitivity and security risks.
  - Choose the appropriate cloud deployment model.

- **Types of Cloud Security Controls**

Cloud security controls are mechanisms designed to protect cloud environments by preventing, detecting, and responding to security threats.

1. Deterrent Controls

Deterrent controls are security measures intended to discourage malicious activities by warning potential attackers of consequences.

- Act as a psychological barrier against attackers.
- Reduce the likelihood of intentional misuse.
- Commonly used to prevent insider threats.

2. Preventive Controls

Preventive controls are security measures that block or reduce vulnerabilities to stop attacks before they happen.

- Actively prevent unauthorized access.
- Strengthen the security posture of cloud systems.
- Help eliminate known security weaknesses.

3. Detective Controls

Detective controls are mechanisms used to identify and detect security incidents in real time or after they occur.

- Monitor cloud environments continuously.
- Detect unauthorized access and abnormal behaviour.
- Help in early identification of security breaches.

4. Corrective Controls

Corrective controls are security measures used to minimize damage and restore systems after a security incident.

- Focus on recovery and damage control.
- Help return systems to a secure state.
- Reduce the impact of security breaches.

❖ Fundamentals Of Computer Security:

Computer security is the process of protecting computers, data, networks, software, and hardware from unauthorized access, misuse, theft, and damage. Although the Internet has made our lives easier, it has also increased risks such as hacking, virus attacks, data theft, and system damage.

As technology is rapidly growing and widely used, hackers and attackers try to harm systems for money, personal gain, or other malicious purposes. Therefore, computer security is essential to protect systems and information from these threats.

<u>Objectives of Computer Security:</u>

Computer security is mainly based on three important objectives: confidentiality, integrity, and availability. These objectives ensure that data is protected, accurate, and accessible when needed.

Confidentiality:

Confidentiality means protecting information from unauthorized access and ensuring that only authorized users can view it. It helps in maintaining the privacy of sensitive data such as personal details, passwords, and business information. If confidentiality is lost, it can lead to data leakage or exposure of confidential information.

It includes data confidentiality, which ensures that private information is not disclosed to unauthorized individuals, and privacy, which gives individuals control over how their personal data is collected, stored, and shared.

Integrity:

Integrity ensures that information remains accurate, consistent, and trustworthy. It prevents unauthorized modification, deletion, or tampering of data. If integrity is lost, the data may become incorrect or unreliable, which can lead to serious issues in systems like banking or healthcare.

It includes data integrity, which ensures that stored and transmitted data is changed only in an authorized way, and system integrity, which ensures that the system functions properly without any unauthorized manipulation.

Availability:

Availability ensures that information and systems are accessible to authorized users whenever required. It guarantees timely and reliable access to data and services without interruptions. If availability is lost, users may not be able to access important information or systems when needed.

This is achieved through proper system maintenance, backups, and protection against attacks like denial of service.

Types of Computer Security

Computer security can be classified into several types based on what is being protected. The main types are explained below:

Cyber Security

Cyber security refers to protecting computers, electronic devices, networks, programs, and systems from cyber-attacks that occur over the internet. These attacks include hacking, malware, phishing, and denial of service attacks. It mainly focuses on securing systems that are connected to online environments.

Information Security

Information security focuses on protecting data and information from unauthorized access, theft, misuse, or damage. It ensures that sensitive information remains safe and is accessed only by authorized users. It is based on

the three main objectives: confidentiality, integrity, and availability of information.

Application Security

Application security involves securing software applications and their data from threats and vulnerabilities. It ensures that applications are designed and maintained in a secure way so that they cannot be easily hacked. It also protects user data and application databases from unauthorized access.

Network Security

Network security focuses on protecting computer networks and the data transmitted over them. It prevents unauthorized access and protects against attacks such as sniffing, spoofing, man-in-the-middle attacks, and other network-based threats. It ensures safe communication between devices.

Endpoint Security

Endpoint security refers to protecting individual devices such as computers, laptops, and mobile phones that connect to a network. These devices can be entry points for attackers, so they are secured using antivirus software, firewalls, and security policies.

Cloud Security

Cloud security focuses on protecting data, applications, and services stored in cloud environments. It ensures that cloud data is secure from unauthorized access, data breaches, and loss, while also maintaining privacy and availability.

Database Security

Database security involves protecting databases from unauthorized access, misuse, or attacks. It ensures that stored data remains safe, accurate, and accessible only to authorized users.

Steps to Ensure Computer Security:

1. Keep the operating system updated regularly because updates fix security weaknesses and protect the system from new viruses and cyber-attacks.
2. Always use a secure and trusted network connection, and avoid using public Wi-Fi, as attackers can easily access data in unsecured networks.

3. Install antivirus software and keep it updated so that it can detect, prevent, and remove viruses and malware from the system.
4. Enable a firewall to monitor incoming and outgoing network traffic and block unauthorized access to the system.
5. Use strong passwords that include uppercase, lowercase, numbers, and special characters, and change them regularly to improve security.
6. Do not trust unknown people or websites, and avoid sharing personal or sensitive information without verification.
7. Keep personal information private and do not share details like passwords, bank information, or personal data on public platforms.
8. Avoid opening email attachments or links from unknown sources, as they may contain malware that can harm the system.
9. Perform online transactions only on trusted and secure websites to prevent financial data theft.
10. Learn basic computer security practices and stay aware of common cyber threats to reduce risks.
11. If a cyber-attack occurs, report it immediately to the cyber cell or authorities for proper action.
12. Avoid pirated software or content, as it may contain viruses or malicious programs that can damage the system.

❖ Cloud Security Architecture:

Cloud Security Architecture is the structured framework (blueprint) used to protect cloud data, applications, and infrastructure using multiple security mechanisms like access control, encryption, and monitoring.

Objectives:

- Protect data confidentiality, integrity, availability
- Prevent unauthorized access and cyber attacks
- Ensure compliance with security standards
- Provide secure and reliable cloud services

Core Components of Cloud Security Architecture

1. Identity and Access Management (IAM)

- Controls who can access what
- **Key techniques:**
  - Multi-Factor Authentication (MFA)
  - Role-Based Access Control (RBAC)
  - Least Privilege Principle
    - Prevents unauthorized access

IAM ensures that only authorized users can access cloud resources. By using MFA, RBAC, and least privilege, it reduces the risk of unauthorized access and insider misuse.

2. Data Security

- Protects sensitive data from leaks
- **Techniques:**
  - Encryption (AES-256, TLS)
  - Data Loss Prevention (DLP)
- **Types:**
  - Data at Rest
  - Data in Transit

Data security protects important information by converting it into encrypted form. Even if attackers access the data, they cannot understand it. DLP helps prevent accidental or intentional data leakage.

3. Network Security

- Protects communication between systems
- **Tools:**
  - Firewalls
  - VPN
  - Intrusion Detection/Prevention Systems (IDS/IPS)
  - Zero Trust Architecture

Network security ensures safe data transfer between users and cloud systems. It blocks unauthorized traffic and detects suspicious activities using tools like firewalls and IDS/IPS.

4. Application Security

- Secures cloud applications from vulnerabilities
- **Methods:**
  - Secure coding practices
  - Regular updates & patching
  - API security

Application security focuses on protecting applications from attacks. Secure coding and regular updates reduce vulnerabilities, while API security prevents unauthorized access through application interfaces.

5. Monitoring & Incident Response

- Detects and responds to threats in real time
- **Tools:**
  - SIEM systems
- **Actions:**
  - Log analysis
  - Alerts
  - Incident handling

This component continuously monitors system activities and quickly responds to threats. It helps in detecting attacks early and minimizing damage through alerts and proper incident handling.

6. Compliance & Governance

- Ensures legal and regulatory compliance
- **Standards:**
  - GDPR, HIPAA, PCI DSS
- **Includes:**

- Audits
- Policies

Compliance ensures that cloud systems follow legal standards and security policies. Regular audits help maintain security and avoid penalties.

Principles of Cloud Security Architecture

1. Defense in Depth

- Multiple layers of security
  - If one fails, others protect

  Using multiple security layers ensures stronger protection, so even if one layer is compromised, other layers continue to secure the system.

2. Least Privilege

- Give minimum access required

Users are given only necessary permissions, reducing risks of misuse and limiting the impact of attacks.

3. CIA Triad

- Confidentiality → Data is private
- Integrity → Data is accurate
- Availability → Data is accessible

The CIA triad is the foundation of security. It ensures data privacy, correctness, and availability at all times.

4. Zero Trust Model

- "Never trust, always verify"
- Every user/device must be authenticated

Zero Trust requires verification for every access request, even from inside the network, improving overall security.

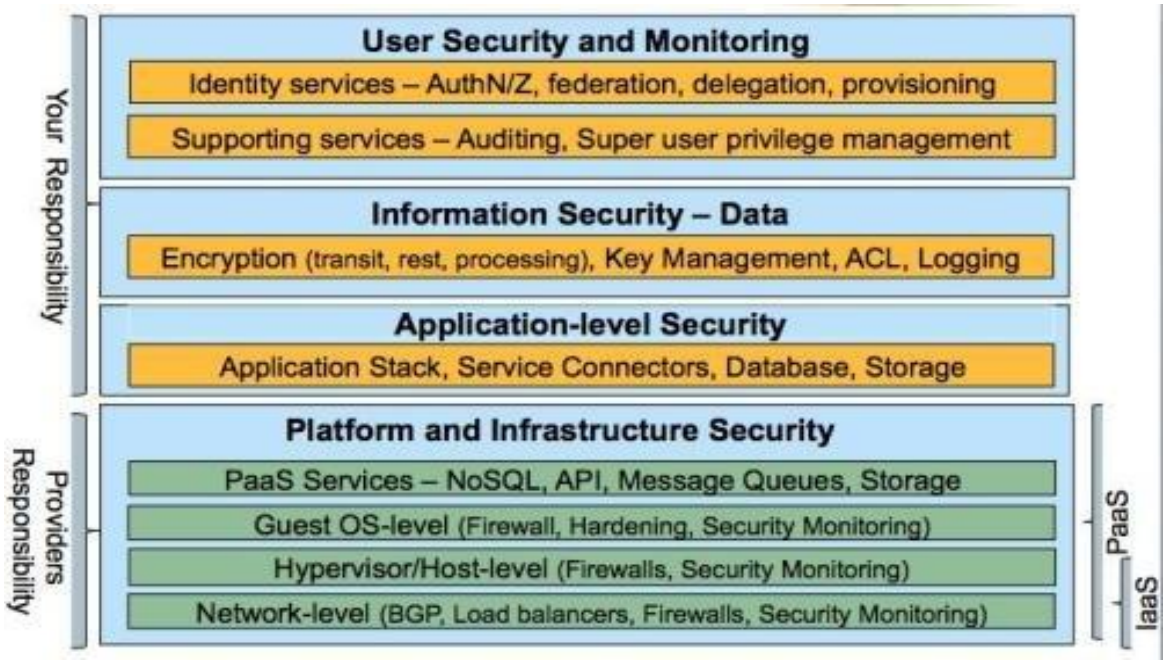
5. Resilience & Backup

- System should recover from failures
- Use backups and disaster recovery

Resilience ensures system continuity during failures. Backups help restore data quickly after attacks or crashes.

Shared Responsibility Model

Cloud security follows a **Shared Responsibility Model**, where **both the cloud provider and the user are responsible for security**, but their responsibilities differ based on the service model.



Models and Responsibilities

1. IaaS (Infrastructure as a Service)

- **Provider Responsibility:**
  - Hardware, servers, networking, data centers
- **User Responsibility:**
  - Operating system, applications, data, firewalls

In IaaS, the provider gives basic infrastructure, but the user has full control over software. So, the user must secure OS, install patches, and manage access. If a vulnerability occurs in the OS, it is the user's responsibility.

2. PaaS (Platform as a Service)

- **Provider Responsibility:**
  - Platform, runtime, servers, OS
- **User Responsibility:**
  - Applications, data, configurations

Here, the provider manages the platform and environment. The user focuses only on application development and data security. If the application has bugs or insecure code, the user is responsible.

3. SaaS (Software as a Service)

- **Provider Responsibility:**
  - Full system (application, infrastructure, updates)

- **User Responsibility:**
  - User access, passwords, usage

In SaaS, everything is managed by the provider. The user only needs to manage login credentials and access control. Weak passwords or misuse of accounts becomes the user's responsibility.

Common Cloud Security Threats

1. Misconfiguration

- Open storage or wrong settings

Improper configuration (like public S3 buckets) can expose sensitive data. This is one of the most common causes of cloud breaches.

2. Phishing Attacks

- Fake login pages steal credentials

Attackers trick users into entering credentials on fake websites. Once login details are stolen, attackers can access cloud systems.

3. Insecure APIs

- Weak APIs allow unauthorized access

APIs connect different services. If they are not secured with authentication and validation, attackers can exploit them to access or modify data.

4. Insider Threats

- Employees misuse access

Authorized users may intentionally or accidentally misuse data. Lack of monitoring and excessive permissions increase this risk.

5. Ransomware

- Data locked and ransom demanded

Malicious software encrypts data and demands payment to restore access. Without backups, recovery becomes difficult.

Steps to Design Cloud Security Architecture

1. Risk Assessment

- Identify vulnerabilities

Before implementing security, organizations must identify risks like weak passwords, open ports, or unencrypted data. This helps in planning proper security measures.

2. Select Cloud Provider

- AWS, Azure, GCP

Choosing a reliable cloud provider is important because each offers different security tools and features. Understanding their security model is necessary.

3. Implement Security Layers

- IAM, encryption, firewalls

Apply multiple layers of security such as access control, data encryption, and network protection. This ensures strong defense against attacks.

4. Monitoring & Response

- Use SIEM tools

Continuous monitoring helps detect suspicious activities. SIEM tools analyze logs and generate alerts for quick response to threats.

5. Ensure Compliance

- Regular audits

Organizations must follow legal and industry standards. Regular audits help identify security gaps and maintain compliance.

❖ Cloud shared responsibility model

The **Cloud Shared Responsibility Model** is a security framework that defines how responsibilities are divided between the cloud provider and the customer. The provider secures the cloud infrastructure, while the customer secures their data and applications.

Core Concept

- Security is **shared between provider and customer**
- Cloud provider → **Security of the cloud**
- Customer → **Security in the cloud**
- Applies to platforms like Amazon Web Services, Microsoft Azure, Google Cloud Platform

Why is it Needed?

- Cloud does not transfer full security responsibility
- Prevents confusion between roles
- Helps in risk management and data protection
- Ensures accountability for security issues

Cloud Provider Responsibility

- Secures physical data centers and hardware
- Maintains servers, storage, and networking
- Protects virtualization layer
- Ensures infrastructure availability and reliability
- Provides compliance and security standards
- Secures underlying services (e.g., Amazon S3, Amazon RDS)

Customer Responsibility

- Protects data (encryption, backup)
- Manages user access and permissions (IAM)
- Secures applications and code
- Configures network settings (firewalls, ports)
- Prevents misconfigurations

Example

Using Amazon Web Services

- *Cloud Provider* takes care of servers and infrastructure
  Example: If a server breaks, AWS fixes it
- *Customer* takes care of data and security settings
  Example: If data leaks due to wrong permissions, it is your mistake

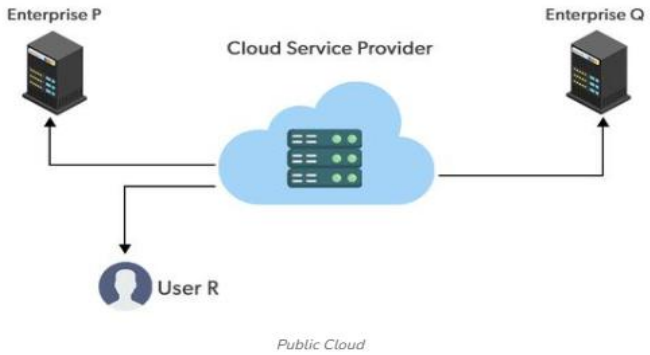
❖ Cloud Deployment Model – Definition

A Cloud Deployment Model defines where the cloud infrastructure is located, who owns and manages it, and how it is accessed. It determines the level of security, control, scalability, and cost of cloud services.

Security in Cloud Deployment Models

1. Public Cloud

The Public Cloud is a cloud environment where infrastructure is owned and managed by third-party providers and delivered over the internet to multiple users (multi-tenant).



Examples:

- Amazon Web Services
- Microsoft Azure
- Google Cloud Platform

Security

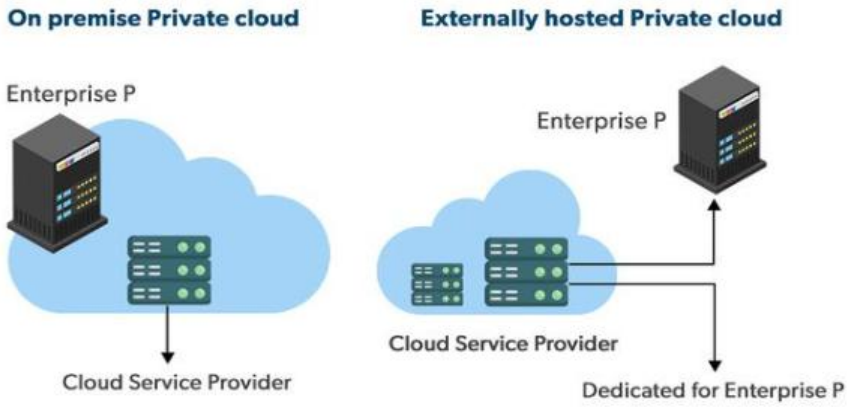
- Shared Responsibility Model - Provider secures infrastructure and User secures data and access
- Data Encryption - Encryption at rest and in transit
- Identity & Access Management (IAM) - Role-based access control & multi-factor authentication
- Network Security - Firewalls, intrusion detection systems

Security Issues

- Multi-tenancy risks (shared resources)
- Data breaches due to misconfiguration
- Limited control over security

2. Private Cloud

The Private Cloud is a cloud environment dedicated to a single organization, providing exclusive access and control over resources.



Security

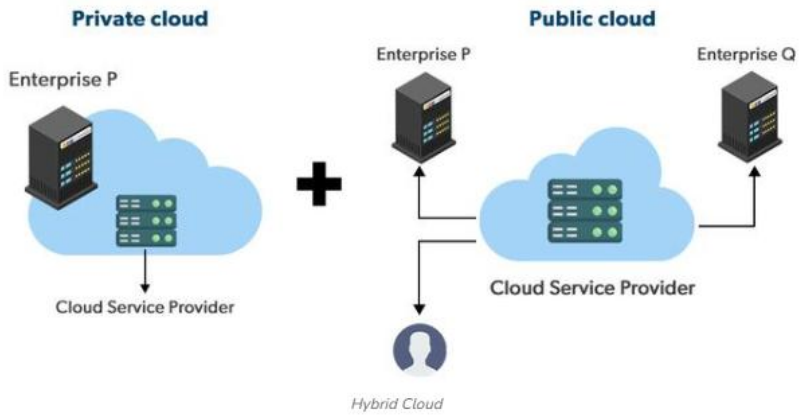
- Full Control - Organization manages entire infrastructure
- Strong Data Privacy - Data is isolated (no sharing)
- Custom Security Policies - Organization defines security rules
- Network Protection - Firewalls, internal security systems

Security Issues

- Depends on internal team expertise
- Misconfigurations can still occur
- High cost for maintaining security

3. Hybrid Cloud

The Hybrid Cloud combines public and private clouds, allowing data and applications to move between them.



Security

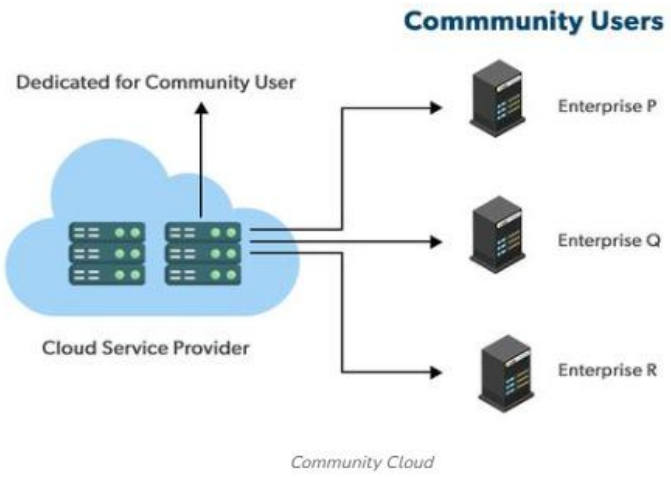
- Data Segmentation
  - Sensitive data → Private cloud
  - Non-sensitive → Public cloud
- Secure Communication - VPNs and encrypted connections
- Unified Identity Management - Centralized access control
- End-to-End Encryption - Protects data during transfer

Security Issues

- Complex security management
- Risk during data transfer
- Inconsistent policies across clouds

4. Community Cloud

The Community Cloud is shared by multiple organizations with common security, compliance, or operational requirements.



Security

- Shared Security Policies - Common standards for all organizations
- Access Control - Restricted to community members
- Compliance-Based Security - Meets industry regulations

Security Issues

- Trust issues between organizations
- Data privacy concerns
- Governance complexity

5. Multi-Cloud

The multi-cloud approach uses multiple public cloud providers simultaneously for different services.

Security

- Provider-Specific Security Tools - Each cloud has its own security features
- Redundancy & Availability - Reduces risk of downtime
- Traffic Routing Security - Direct users to secure locations

Security Issues

- Complex security management
- Difficult to maintain consistent policies
- Increased attack surface